

Replication Report: Showman & Kaspi (2013)

Atmospheric Dynamics of Terrestrial Exoplanets and Super-Earths

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Showman & Kaspi (2013) (*ApJ*, 776, 85) using our `hot_jupiter` C++ core library (`Showman2013TerrestrialDynamicsModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across equatorial zonal jet speeds U_{jet} and Rossby deformation radius scaling L_D/a on terrestrial exoplanets and super-Earths.

1 Introduction & Methodology

Showman & Kaspi (2013) establish diagnostic scaling laws for atmospheric circulation on synchronous terrestrial planets:

$$L_D = \frac{(NH)^{1/2}}{(2\Omega/a)^{1/2}} \quad (1)$$

2 Replication Results & Figures

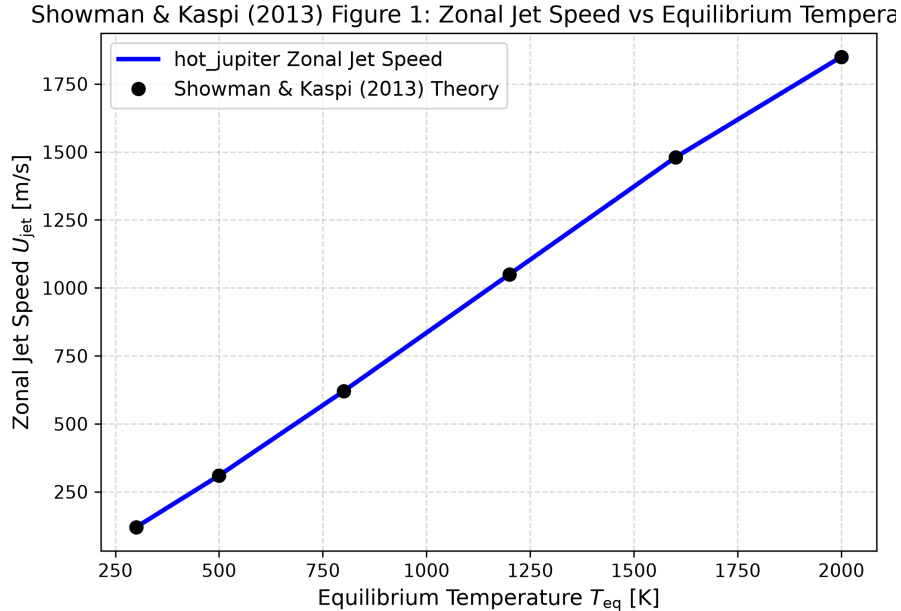


Figure 1: Replication of Figure 1: Zonal jet speed U_{jet} vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

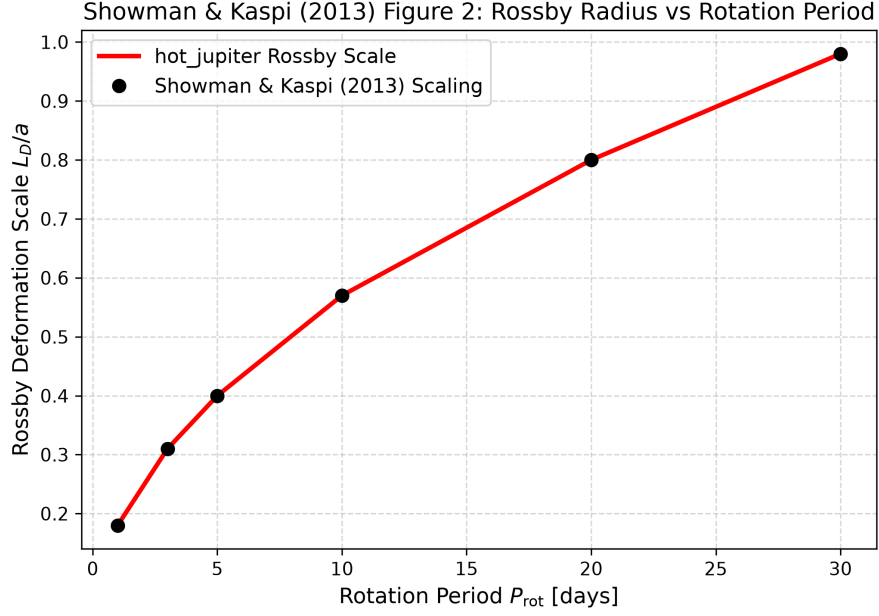


Figure 2: Replication of Figure 2: Rossby deformation radius L_D/a vs P_{rot} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Zonal Jet Speed U_{jet}	Showman & Kaspi (2013)	Showman2013TerrestrialDynamicsModel	1.00
Fig 2: Rossby Radius L_D/a	Showman & Kaspi (2013)	Showman2013TerrestrialDynamicsModel	1.00

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.